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Element C:

PRIORITY #1: ATTACHABLE/DETACHABLE FROM TABLE

The function of our design will be able to have a base that can attach and detach from a table. The repairability of our design should be able to attach to the table via velcro or a supporting base nailed into the table so that the actual container doesn't move. Our goal for the reliability of the base is that it is easy to take off the desk for moving around, but still secure so nobody can take the calculators. The aesthetic of our design is simple, with not too much going on to make it look visually appealing. Our design will be durable enough so that students can't just take the crate off of the base. The base will keep the actual design on the table while it is attached to it. The productibility of the product should exceed our expectations, with the base being frequently used through the day. Our design's simplicity will be based on how we plan to keep the base on the table, and the actual design on the base. Since our overall design will be able to be carried around and not have to be stuck on the desk forever, I'd say that it is pretty compact and travel-worthy.

PRIORITY #2: RECYCLABLE MATERIALS

As our community partner stated, a big requirement for him is the use of recyclable materials for this project. Our previous research concluded that plastic bottles, beverage cans, paper, cardboard, food containers, and glass bottles can be recycled as long as they are clean. Due to recent research, it was found that 3D printers also use recyclable materials. This would be ideal since it would have more durability and have a higher quality than trying to put together different materials. Along with durability, combining different materials would also result in the product looking messy and unprofessional which wouldn't satisfy our community partner. The 3D printing will be originated on TinkerCad since the engineers are experienced in that field.

PRIORITY #3: COMPACT & ASTECHIC

Requirements: The shape can be about 1ftx4inx6in (this will be updated after we get the measurements of his calculators). We will 3d print the design to be compact and not consume too much space on a desk. The calculator will be partially visible, and will fit snugly within the holder. The holder will have a handle so the calculators can be easy to move. The engineers want the design to be discrete and as modern as possible to go with the astechic. The stakeholder would like something that would blend in to their work environment, and not be too bulky. This is why the design must be compact. Since the design will be 3d Printed, we must make sure that the design in TinkerCad is perfected. Also, engineers will be able to control the design of the holder with more precision. The primary stakeholder, Mr. Scott, would prefer that the design be easy to find, so engineers will find a way to make the design stand out in a way that doesn't ruin the design.

PRIORITY #4: QUALITY

The quality is very important to the design as we need something that can last for a while not something that will last for at most a few days. We also need to make sure it is durable enough to not break just in case if it ever drops on the floor. We also have to make sure it can hold a lot of calculators so that it easier to move them. We also should make sure that the calculator holder won't fall off so it needs to be well balanced. We also need the calculator holder to not wear down easily. If it just just falls apart after a few days than it will just be a massive waster of time. We want to make it so the calcultor holder lasts for at least a few months.